

2024 Midterm 1: Question 4 (20 points)

Consider the following Dafny program that inefficiently calculates the cube of a number:

```
method Cube(m:int) returns (y:int)
requires m > 0
ensures y == m*m*m
{
  y := 0;
  var x := m * m;
  var z := m;
  while (z > 0)
  {
    z := z - 1;
    y := y + x;
  }
}
```

Fill in the annotations below to show that the Hoare triple given by the outermost pre- and postconditions is valid. Be completely precise and pedantic in applying Hoare rules: write assertions in exactly the form given by the rules, rather than just logically equivalent ones. Working backward from the end should require the rule of consequence only at the places marked $\rightarrow$. Those implication steps may silently rely on the usual rules of arithmetic.

Edit the nine answer commands near the top of this file. Labels A–I are worksheet labels, not part of the assertions. The original decorated listing uses uppercase Z in its guard; that spelling is preserved below. The method above uses lowercase z .

- $\{\{ m > 0 \}\}$
 $\rightarrow$
- A $\{\{ (0 = (m - m) * (m * m)) \wedge (m * m = m * m) \wedge (m \geq 0) \}\}$
 $y := 0;$
- B $\{\{ (y = (m - m) * (m * m)) \wedge (m * m = m * m) \wedge (m \geq 0) \}\}$
 $\text{var } x := m * m;$
- C $\{\{ (y = (m - m) * x) \wedge (x = m * m) \wedge (m \geq 0) \}\}$
 $\text{var } z := m;$
- D $\{\{ (y = (m - z) * x) \wedge (x = m * m) \wedge (z \geq 0) \}\}$
 $\text{while } (Z > 0) \{$
- E $\{\{ ((y = (m - z) * x) \wedge (x = m * m) \wedge (z \geq 0)) \wedge (z > 0) \}\}$
 $\rightarrow$
- F $\{\{ ((y + x) = (m - (z - 1)) * x) \wedge (x = m * m) \wedge ((z - 1) \geq 0) \}\}$
 $z := z - 1;$
- G $\{\{ ((y + x) = (m - z) * x) \wedge (x = m * m) \wedge (z \geq 0) \}\}$
 $y := y + x;$
- H $\{\{ (y = (m - z) * x) \wedge (x = m * m) \wedge (z \geq 0) \}\}$
 $\}$
- I $\{\{ (y = (m - z) * x) \wedge (x = m * m) \wedge (z \geq 0) \wedge \neg(z > 0) \}\}$
 $\rightarrow$
 $\{\{ y = m * m * m \}\}$